Hall Ticket Number:

MR-22

Code: B324CS4S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July/August-2024

Object Oriented Programming through Java

(Common to CSE, CSB and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

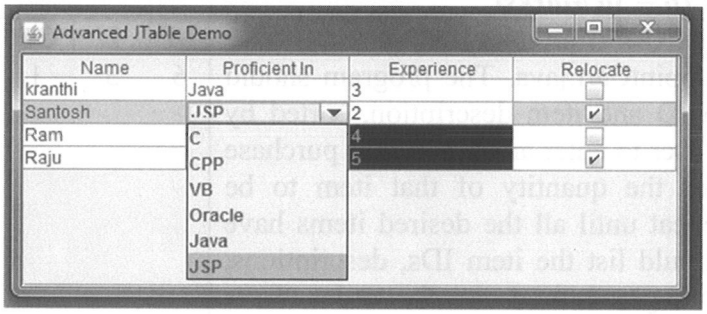
Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Write four major differences between C and Java.	1	2	1	1,2
1. b)	List the java buzzwords.	1	1	1	1,2
1. c)	How do we tell java that we want to use a particular package in a file?	1	2	2	1,2
1. d)	Define the need for static keyword.	1	2	2	1,2
1. e)	List the Common java Exceptions.	1	1	3	1,2
1. f)	How do we start a thread?	1	1	3	1,2
1. g)	What is event handling mechanism? Give different event handling classes.	1	2	4	1,2
1. h)	Define layout manager and list different layout managers.	1	2	4	1,2
1. i)	How many arguments can be passed to an applet using <PARAM> tags?	1	3	5	1,2
1. j)	List the components of MVC architecture.	1	1	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Create a Java program called ShoppintCart.java. The program should display a list of items (with item ID and item description, sorted by item ID) and their prices. Ask the user to enter an item ID for purchase (or-999 to quit) and also ask for the quantity of that item to be purchases. This process should repeat until all the desired items have been selected. The final output should list the item IDs, descriptions, quantities, and subtotals, with the grand total for the order. User currency formatting for all currency values.	6	3	1	1,2
2. b)	Explain the features and benefits of object oriented programming.	4	2	1	1,2
	OR				
2. c)	Write a java program to simulate the operations of numerical calculator to perform the functions Addition (+), subtraction (-), Multiplication (*), Division (/).	6	3	1	1,2
2. d)	Briefly discuss about the control statements used in java.	4	2	1	1,2
3. a)	Identify the differences between method overloading and overriding with examples.	5	3	2	1,2
3. b)	What is the use of package? Write a java program to explain the concept of package.	5	2	2	1,2
	OR				
3. c)	Describe the various forms of implementing interfaces. Give examples of java code for each case.	5	3	2	1,2
3. d)	Write short notes on Abstract classes and methods.	5	2	2	1,2
4. a)	Explain the concept of user defined exception with suitable example.	5	3	3	1,2
4. b)	What is synchronization? Explain with an example.	5	2	3	1,2
	OR				

P.T.O.

4. c)	What is a **finally** block? When and how is it used? Give a suitable example.	5	2	3	1,2
4. d)	Write a program to create threads where one thread prints 1 to 100 and another thread prints 1 to 200.	5	3	3	1,2
5. a)	Discuss about the AWT component classes with suitable example program.	5	2	4	1,2,3
5. b)	Explain the steps involved in creation and handling of menus.	5	2	4	1,2
	OR				
5. c)	Write a program for creating GUI for reserving bus ticket, store the passenger information in file. Give concession for senior citizen, child and army people. Keep seat reserved for privileged people till one hour before the departure time. Use exception for the invalid date, name. Amount. Ticket id must be generated automatically. Only once name of the bus should be stored in memory.	5	4	4	1,2,3
5. d)	Explain the concept of event delegation model.	5	2	4	1,2
6. a)	Write a program to create a simple login page with two labels, two text fields and one button with swings.	5	4	5	1,2,3
6. b)	Describe the different stages in the life cycle of an applet. Distinguish between **init ()** and **start( )** methods.	5	2	5	1,2
	OR				
6. c)	Write a simple JAVA Program to create the following window and store the entered data in the file. 	5	4	5	1,2,3
6. d)	Discuss the steps involved in developing and running a local applet.	5	2	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome